



Instructor: Date:

Learner: Regional Branch:

Program: Center:

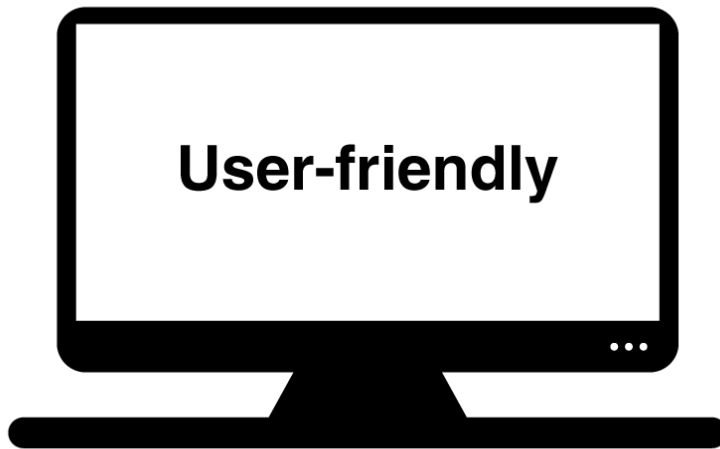
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Let's Start!

Let's design a DBMS!



Learning Outcome:

Exchange information about relational database management system.

Ask your partner:

- How important are smartphone apps for you? Why?
- Which ones do you use more frequently? Why?

Can Do Statements:

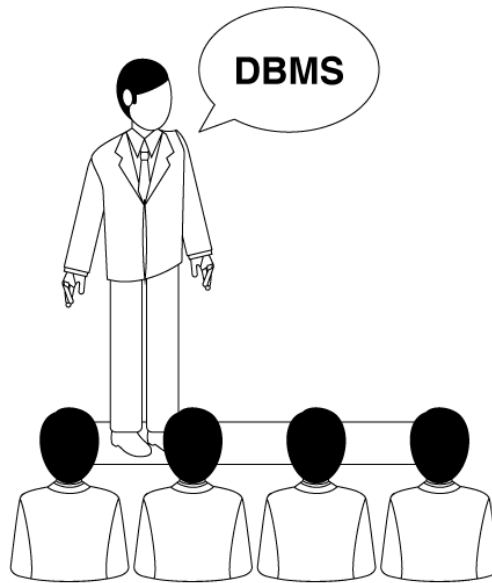
- I can talk about features of the architecture of a DBMS.
- I can explain what three-tier architecture is.
- I can talk about the components of three-tier architecture.





Let's Explore!

A Speech on DBMS Architecture



Instructions: Read the text below.

Let's talk about DBMS! In order to understand how a DBMS can be designed, we need to be clear about its architecture. The first thing we need to decide is whether we want it to be a single tier, a multi-tier or an n-tier one (which splits the system in different linked modules). Just to clarify, a tier is like a level; let's say that it is like designing a house of one, two or three levels. It's the same concept that could be used when designing DBMSs.

Last week, for example, I had a meeting with some coworkers. We had been notified that we needed to start to work on a DBMS so we met to decide how many levels it would have. We thought that a 1-tier design could be seen as an unsuitable resource because we needed other people to be able to access it from different places, which could not be done by using a 1-tier. Then, before we had our coffee we had already decided that a 2-tier architecture could not be perceived as useful either. So we reached to the point that the only option was a three-tier architecture.

However, what does that mean? 3-tier architecture works as follows: The database tier, which is the level at the bottom, holds all the information and query processing languages as well as the restraints for users. Next, we have the second level or the application tier. This tier acts as a mediator, presenting the information from the database to the user. Finally, we have the user tier, which is the front end of the DBMS. End users interact with this application, not knowing the existence of the database level that is residing in a different place. Three-tier architecture is the most flexible one because it provides independency to its three components so, all the components can be manipulated without affecting the others.



**Let's Practice!**

Pam studied really hard for her test!

Instructions: Complete the sentences by writing its corresponding words. Use capital letters where necessary. Use: started working, had understood, She had read, learned, had learned.

1. Pam _____ at the university what a 3-tier architecture meant and implied.
2. Pam _____ how a DBMS could be designed.
3. She had also _____ that a DBMS could be a single tier or a multi-tier one.
4. Pam had _____ on a DBMS before she decided how many levels it would have.
5. _____ that a 1-tier design was not very useful for many users.

**Let's Practice!**

The three tier architecture is simply the best!

Instructions: Read the sentences. Write the missing word(s) in the blanks. Use: she had emphasized on, he had said that, what she had said about, had mentioned, had reminded us that query.

I remember that before our general meeting, Michael _____ some important features about the three tier architecture that we often use in DBMS. For example, _____ this type of architecture was the most flexible one because there was no dependence among its components. Furthermore, I remember that Carly _____ processing languages could be included in the tier level at the bottom and that we could also include user restraints in there. In fact, _____ how the second level, the application tier, worked as a mediator. Finally, I can't remember quite well _____ the user tier, though I believe it was something related to the interaction between the final user and the application.





Extension Activity

Talking About the Architecture of a DBMS!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- As you already know, a DBMS can be seen in three different tiers. Make a brochure explaining each tier.
- Fold a sheet of paper to create your brochure. Use a different side to explain each tier and give relevant examples.
- Search on the web for extra information to complement your ideas.
- Get ready to show your brochure to your classmates and explain your ideas.

Self-Assessment

Yes

No

- | | | |
|--|-----------------------|-----------------------|
| 1. I make a brochure explaining each tier of a DBMS. | ☐ | ☐ |
| 2. I follow the steps to create the brochure and I give relevant examples. | ☐ | ☐ |
| 3. I surf the web to complement my ideas. | ☐ | ☐ |
| 4. I present my final brochure to the class. | ☐ | ☐ |



Target Words

- **architecture** (noun): the rudimental design of a computer hardware or software
e.g. Let's talk about DBMS! In order to understand how a DBMS can be designed, we need to be clear about its architecture.
- **unsuitable** (adjective): not appropriate at all
e.g. We thought that a 1-tier design could be seen as an unsuitable resource because we needed other people to be able to access it from different places, which could not be done by using a 1-tier.
- **reach to the point** (expression): to get to a certain point
e.g. So we reached to the point that the only option was a three-tier architecture.
- **bottom** (noun): the lowest level or position
e.g. The database tier, the level at the bottom, holds all the information and query processing languages as well as the restraints for users.
- **restraint** (noun): the limits or unauthorized actions
e.g. The database tier, the level at the bottom, holds all the information and query processing languages as well as the restraints for users.
- **mediator** (noun): a person or program that brings two parties into an understanding
e.g. This tier acts as a mediator, presenting the information from the database to the user.



Learning Points

Past Perfect

You use the **past perfect** to talk about something that happened before another action in the past.

The **past perfect** refers to two different actions in the past; however, one of them happened before the other. In other words, we can say that the **past perfect** tense refers to 'the past of the past' (a past action that happened before another past action). Now, the action that is closest to the present time uses the past simple tense while the one that happened before that uses the past perfect tense.

The structure that we follow to make sentences using the past perfect tense is:

Subject + auxiliary verb 'had' + action verb in past participle + complement.

We can express both situations (using the past simple and the past perfect) in different ways.

For example:

*I **had saved** my files **before** the computer **crashed**.*

***Before** the computer **crashed**, I **had saved** my files.*

(*The person saved the files before the computer crashed.)

*She **had studied** her notes **before** she **presented** her exam.*

***Before** she **presented** her exam, she **had studied** her notes.* (*The person studied her notes before she presented the exam.)

Finally, remember that the abbreviation of 'had' is just " 'd ".

For example:

***She'd** called me before about that issue.*

***They'd** designed databases before they designed webpages.*

Examples

- *We **had been notified** that we needed to start to work on a DBMS so we met to decide how many levels it would have.*
- *Then, before we had our coffee we **had already decided** that a 2-tier architecture could not be perceived as useful either.*





Learning Points

Three Tier Architecture

We call the **three tier architecture** like that because it separates tiers in 3 parts that can actually be a component of different platforms.

An advantage of using the three tier architecture is that as each part is independent, you can update or replace it without affecting the functioning of the other two parts.

As its name indicates, the three tier architecture has three levels (down-up):

1. The **data tier**, which is where all the information is located.
2. The **logic tier**, which is the level that makes all the hard work processing the information to retrieve it from the database and present it to the user.
3. The **presentation tier**, which is the level that receives the information that is requested by the user to be displayed.

Check the following graph to learn more about the three tier architecture.

